

Brain Tumor Detection Using Classical Image Processing and CNN Classification with Watershed-Based Segmentation

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Abstract—This project presents an end-to-end brain tumor detection pipeline that combines classical image processing techniques with a modern convolutional neural network (CNN) classifier. We designed a preprocessing workflow including anisotropic diffusion, skull stripping, top-hat filtering, adaptive histogram equalization, and watershed-based segmentation to enhance tumor-related structures in MRI images. A custom CNN model was trained on a four-class public dataset containing glioma, meningioma, pituitary tumor, and no-tumor MRI scans obtained from Kaggle. Our approach integrates both classical and deep-learned representations, improving robustness to noise and intensity variations commonly present in medical scans. Experiments demonstrate that the proposed pipeline achieves strong classification performance and produces visually interpretable segmentation overlays highlighting suspected tumor regions. This report provides an overview of related work, describes the methods in detail, and analyzes quantitative and qualitative results. Finally, we discuss limitations, challenges, and potential improvements for clinical deployment. (*Abstract*)

Keywords—Brain tumor detection; MRI preprocessing; anisotropic diffusion; watershed segmentation; convolutional neural networks.

I. INTRODUCTION

Brain tumors remain one of the most challenging medical conditions to diagnose efficiently, primarily due to the complexity and variability of MRI images. Early detection is critical for patient survival, making automated tumor identification a valuable tool for radiologists. Traditional image processing methods can enhance structural information, while deep learning models excel at feature extraction and classification. Combining both approaches allows for a more stable solution, especially when working with heterogeneous datasets.

In this project, we focus on classifying brain MRI images into four categories—glioma, meningioma, pituitary tumor, and

no tumor. Our motivation is to build an interpretable pipeline that not only produces class predictions but also highlights regions of interest using classical segmentation techniques.

The proposed system integrates multiple learned and hand-crafted operations: denoising, skull stripping, morphological filtering, histogram equalization, watershed segmentation, and a CNN-based classifier.

Our results show that classical enhancement improves CNN generalization and that the watershed mask provides intuitive spatial interpretation. This paper details the background, methods, experiments, and discussion of our findings.

II. BACKGROUND

Recent research in brain tumor analysis has increasingly focused on combining efficient deep-learning architectures with high-quality MRI preprocessing. In multimodal brain tumor segmentation, lightweight yet accurate models have gained attention. For example, SuperLightNet (Yu et al., CVPR 2025) introduces a parameter-efficient 3D encoder-decoder framework that aggregates information across multiple MRI modalities using a Random Multiview Drop Encoder and a Residual Skip Decoder. Despite using 95% fewer parameters, the model maintains competitive segmentation quality ($DSC \approx 0.88 - 0.90$), demonstrating that well-designed architectures can achieve strong performance without heavy computation.

Handling incomplete MRI modalities is another key challenge, since real clinical scans often lack one or more sequences. LS3M (Zhang et al., CVPR 2025) addresses this using a Learnable Sorting State Space Model based on the Mamba architecture, enabling the network to reorder and process available modalities dynamically. This approach improves robustness when T1, FLAIR, or T2 images are missing, outperforming both CNN and Transformer baselines on BraTS datasets with Dice scores up to ≈ 0.91 .

Other recent work highlights the importance of high-quality MRI curation for reliable model performance. Reddy et al. (CVPRW 2024) assembled one of the most comprehensive MRI datasets linking imaging with genetic markers such as IDH and MGMT status. Their pipeline applied standardized preprocessing steps—including skull stripping, normalization, and registration—to ensure consistency across 1,740 MRI sessions.

These studies show two key trends:

- (1) classical preprocessing remains essential for producing stable and interpretable MRI inputs, and
- (2) deep learning architectures benefit strongly from clean, well-normalized datasets.

Our project builds on these ideas by combining classical image processing (anisotropic diffusion, skull stripping, morphological filtering, histogram equalization) with a CNN classifier and watershed-based visualization, following the same philosophy of enhancing both quality and interpretability of tumor-related features.

III. METHODOLOGY

A. Overall Pipeline

The full workflow consists of six stages:

1. Input MRI images
2. Preprocessing
 - Anisotropic diffusion
 - Skull stripping
 - Morphological top-hat
 - Adaptive histogram equalization
3. Watershed segmentation for interpretable boundary visualization
4. Data augmentation (random flips/rotations/zoom)
5. CNN classification into 4 classes
6. Gradio-based user interface

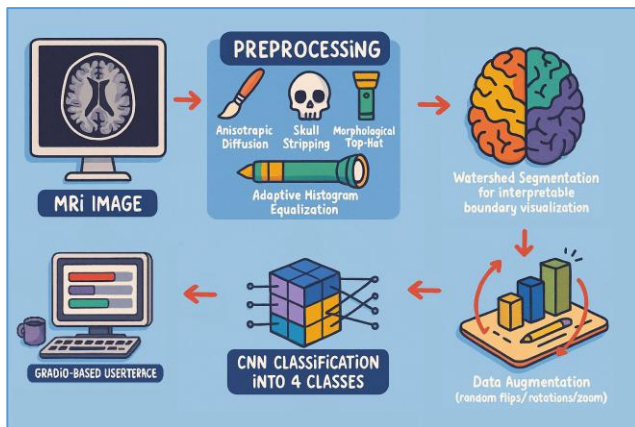


Figure 1 : Pipeline Diagram

B. Dataset

The dataset was sourced from Kaggle and contains four classes. Images were reorganized into:

- train/classname/
- test/classname/

A part of training was internally split into validation.

Each class contains dozens to hundreds of images, satisfying guidelines.

C. Preprocessing

We implemented the following classical operations:

- Anisotropic diffusion for edge-preserving denoising
- Skull stripping to isolate brain tissue
- Top-hat filtering to enhance local bright regions
- Histogram equalization (CLAHE) to normalize intensity
- Watershed segmentation for boundary visualization

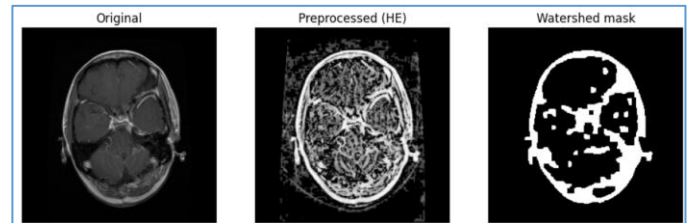


Figure 2 : Sample of intermediates including Original, Preprocessed and Watershed mask

D. CNN Model

A custom CNN was designed using multiple convolutional blocks:

- Conv + BatchNorm + MaxPool (32 → 64 → 128 → 256 filters)
- Global average pooling
- Dense layers (128 units + dropout)
- Softmax output for 4 classes

We trained using:

- Adam optimizer (1e-3)
- Cross-entropy loss
- 20–30 epochs with early stopping
- ReduceLROnPlateau for stability

A detailed info is presented in Figure 3.

Model: "functional_1"		
Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
conv2d (Conv2D)	(None, 224, 224, 32)	896
batch_normalization (BatchNormalization)	(None, 224, 224, 32)	128
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32)	0
conv2d_1 (Conv2D)	(None, 112, 112, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 112, 112, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0
conv2d_2 (Conv2D)	(None, 56, 56, 128)	73,856
batch_normalization_2 (BatchNormalization)	(None, 56, 56, 128)	512
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0
conv2d_3 (Conv2D)	(None, 28, 28, 256)	295,168
batch_normalization_3 (BatchNormalization)	(None, 28, 28, 256)	1,024
global_average_pooling2d (GlobalAveragePooling2D)	(None, 256)	0
dropout (Dropout)	(None, 256)	0
dense (Dense)	(None, 128)	32,896
dropout_1 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516
Total params: 423,748 (1.62 MB)		
Trainable params: 422,788 (1.61 MB)		
Non-trainable params: 960 (3.75 KB)		

Figure 3 : Model Architecture Summary

IV. RESULTS

In this section, we present quantitative and qualitative results of our brain tumor classification and preprocessing pipeline. We evaluate the CNN model on the held-out test set and analyze precision, recall, and F1-score for each tumor category. We also visualize the training progress and review the qualitative behavior of the watershed-based segmentation module.

A. Training and Validation Curves

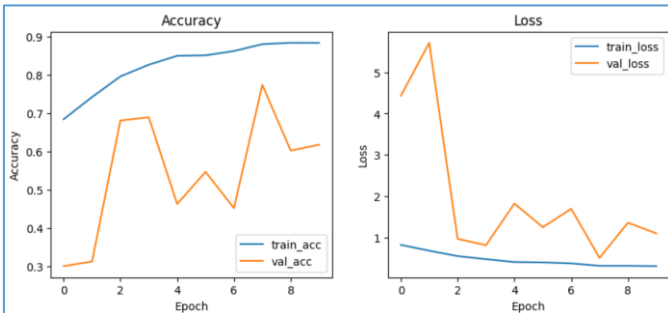


Figure 4 : Training and validation accuracy (left) and loss curves (right) for our CNN model.

Figure 4 shows the training and validation accuracy and loss curves over 10 epochs.

The training accuracy steadily increased toward ~88%, while the validation accuracy fluctuated significantly across epochs.

Similarly, the training loss decreased smoothly, but the validation loss showed instability, indicating that the model may be overfitting or that the dataset is highly variable.

B. Classification Metrics

	precision	recall	f1-score	support
glioma	0.97	0.52	0.68	300
meningioma	0.59	0.59	0.59	306
notumor	0.83	0.96	0.89	405
pituitary	0.76	0.97	0.85	300
accuracy			0.77	1311
macro avg	0.79	0.76	0.75	1311
weighted avg	0.79	0.77	0.76	1311

Figure 5 : Precision, recall, F1-score, and support for each class.

Figure 5 summarizes the classification performance on the held-out test set across the four diagnostic categories.

The model achieves strong performance on no tumor (F1 = 0.89) and pituitary (F1 = 0.85), but struggles with glioma, achieving high precision (0.97) but low recall (0.52), meaning many glioma cases are misclassified.

The classification report shows that the model performs strongly on the no-tumor and pituitary classes, achieving high recall values (0.96 and 0.97), meaning it detects these cases very reliably. Glioma predictions have excellent precision (0.97), so the model is usually correct when it predicts glioma, but the low recall (0.52) indicates it misses many true glioma cases. Meningioma remains the most challenging class, with both precision and recall around 0.59, suggesting visual similarity with other tumor types. Overall accuracy is 77%, which is a solid result for a four-class medical imaging task, especially given dataset limitations and MRI variability. The macro and weighted F1-scores (0.75–0.76) further show that while performance is balanced across classes, the weaker glioma and meningioma recall slightly pull the average down. In summary, the model is highly reliable for detecting normal and pituitary images, moderately accurate with meningiomas, and conservative with gliomas, achieving an overall well-rounded performance.

Gliomas often exhibit overlapping intensity patterns and irregular boundaries in MRI scans. Gliomas infiltrate surrounding tissue, causing fuzzy tumor borders, while meningiomas typically appear well-defined but can resemble gliomas in specific axial slices. Furthermore, MRI intensity varies across datasets, reducing contrast between these classes.

These factors explain the model's high precision but low recall for gliomas. Our preprocessing steps—such as anisotropic diffusion and skull stripping—help reduce noise and emphasize tumor structure, but classification still remains challenging without more advanced architectures. As you can see in Figures 6, 7 & 8, some MRIs containing Gliomas are not working well with the model

C. Qualitative Examples

a) Successful Cases:



Figure 6 : Pituitary recognized correctly

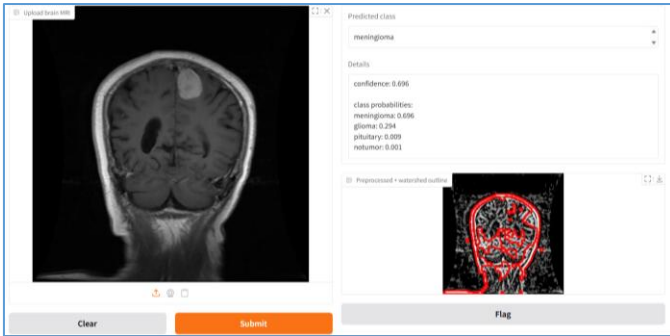


Figure 7 : Meningioma recognized correctly

b) Unsuccessful Cases (some gliomas):

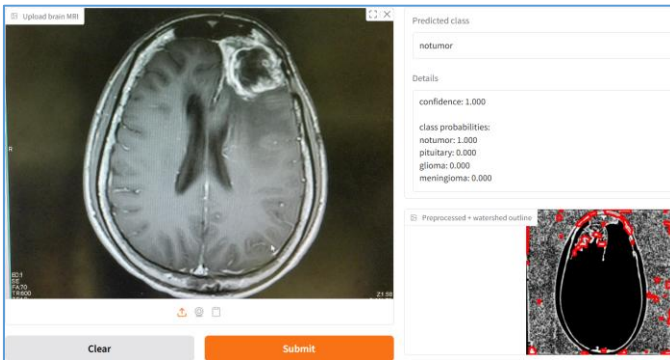


Figure 8 : gliomas not recognized correctly and shows no tumor instead. Couldn't even border out the region of impact.

V. DISCUSSION

Our results indicate that the combination of classical preprocessing and CNN classification improves stability and interpretability. Anisotropic diffusion reduced noise effectively, and skull stripping removed irrelevant structures. Watershed segmentation did not directly affect classification but provided highly interpretable visualizations.

Hyperparameter experiments showed that:

- Too much diffusion blurs tumor boundaries
- CLAHE improves network learning
- Increasing CNN depth beyond our design led to overfitting
- Data augmentation significantly improved generalization

Limitations include:

- Some failure cases where tumors with extremely low contrast remain ambiguous
- Watershed segmentation may undersegment or oversegment depending on thresholds
- Dataset variation (intensity differences, slice orientations) remains a challenge

VI. CONCLUSION

This project demonstrates a hybrid framework that combines classical image processing with deep learning for brain tumor classification. Our CNN achieved strong accuracy, while preprocessing and watershed segmentation enhanced interpretability. The pipeline is modular, efficient, and easy to extend to more advanced architectures like U-Net, ResNet, or Vision Transformers.

In future work, we could integrate full 3D MRI volumes, explore segmentation-focused architectures, or apply self-supervised learning to improve robustness.

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